A CNN Framework With Slow-Fast Band Selection and Feature Fusion Grouping for Hyperspectral Image Change Detection

Xianfeng Ou, *Member, IEEE*, Liangzhen Liu, Bing Tu, *Member, IEEE*, Guoyun Zhang, *Member, IEEE*, and Zhi Xu

***Abstract*— Change detection approaches can detect changed areas of the same scene at different times. Hyperspectral remote-sensing images contain large amounts of spectral information at high resolution. As hyperspectral datasets become abundant, more and more change detection technologies use hyperspectral images as raw data. Hyperspectral images suffer from band redundancy. There is an urgent need to improve the directionality of change of information features. To solve these problems, in this article, we propose a CNN framework involving slow-fast band selection (SFBS) and feature fusion grouping (SFBS-FFGNET) for hyperspectral image change detection. The main contributions of this article are as follows: 1) based on slow feature analysis (SFA), an SFBS method is proposed, which selects slow and fast feature bands to better extract changed and unchanged features, to more effectively separate changed and unchanged pixels; 2) we used a difference matrix to enrich the level of change information to provide more change characteristics for change detection; and 3) an FFG method was used to generate a more discriminative feature group, and the related loss function was designed. Experimental results on multiple real hyperspectral datasets showed that SFBS can reduce the operating load on the computer and improve the accuracy of change detection, and FFG can also improve the accuracy of change detection. In summary, SFBS-FFGNET is superior to most existing change detection methods.**

***Index Terms*— Change detection, difference matrix, feature fusion grouping (FFG), hyperspectral image, slow-fast band selection (SFBS).**

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I. Introduction

HYPERSPECTRAL remote-sensing images have high spectral and spatial resolution, including hundreds of bands, and contain rich spectral information, which greatly enhances the value of hyperspectral remote-sensing images in many applications [1]. The use of hyperspectral images for change detection requires a large number of datasets, and rich datasets can most effectively verify the effectiveness and stability of a change detection method. With the rapid development of aerospace remote-sensing technology in recent years, increasing numbers of satellites have been launched, and the amount of hyperspectral remote-sensing data that can be used for research and analysis has increased. In recent years, hyperspectral remote-sensing images have been widely used in many fields, such as agricultural development, military layout, vegetation ecological inspection, environmental inspection, urban planning, and disaster assessment [2].

Change detection is the process of identifying changed areas in two images at different moments in the same scene. It is usually divided into the following steps.

1) *Hyperspectral Data Preprocessing:* Data preprocessing mainly involves registering the dual-temporal image, and correcting the radiation to eliminate the influence of light and atmosphere on the hyperspectral image [2]. Appropriate preprocessing operations can improve the effectiveness of the data and lay the foundations for subsequent extraction of changed areas [3].
2) *Change Method:* The selection of a change detection method is the most important step in the whole process. The use of a good change detection method determines the accuracy of the final detection [3], [4].
3) *Results and Evaluation:* The result of change detection is usually displayed in the form of two categories: unchanged pixels and changed pixels. Finally, appropriate evaluation methods and indicators for the evaluation of the classification results must be selected [5].

Remote-sensing change detection originated from the image difference method proposed by Weismiller *et al.* [6] in 1977, which took the first step in the remote-sensing change detection. In the initial stage of change detection development, the mainstream methods include algebra [7], transformation [8], [9], and classification [10]. These methods have high requirements for image preprocessing, and they are still based

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on image pixels to do calculations or operations. First, make a difference to the dual-temporal pixel pair, and then set a reasonable threshold to obtain a change detection map, but the change detection accuracy obtained in this way is often not very high. As more and more new methods are proposed, the process of change detection has taken a step forward. The most popular methods are classification and regression trees (CARTs) [11], convolutional neural networks (CNNs) [12], support vector machines (SVMs) [13], and other deep learning and machine learning methods. These methods use more complex mathematical operations to improve the overall accuracy (OA) of change detection classification, but because the resolution of remote-sensing images was not high enough at the time, and the remote-sensing dataset was not rich enough, the detection results were not ideal. As the launch frequency of artificial satellites continues to increase, more high-resolution remote-sensing image databases have been enriched, the development of change detection has reached a new height, and object-based image analysis (OBIA) [14] methods have been applied to the field of change detection. This method has a good performance in visualizing the change detection map and the overall detection accuracy, but it does not incorporate more complex calculations, and it is still lacking in the feature depth of the change information. In recent years, due to the explosion of big data, deep learning methods have become more popular than at any time. Deep learning networks such as deep belief networks (DBNs) [15], CNNs [12], recurrent neural networks (RNNs) [16], convolutional autoencoding (CAE) [17], etc. have been widely used in remote-sensing image change detection, but there will still face the problem of not strong universality and robustness.

Since hyperspectral remote-sensing images can contain large amounts of information, dual-temporal hyperspectral remote-sensing data are an excellent source of data for change detection. However, hyperspectral image change detection technology is still in the developmental stages. The main challenges are summarized as follows.

1) *Hyperspectral Band Redundancy [18]:* The hyperspectral remote-sensing image has rich bands, and different bands have different contributions to change detection. Bands that do not contribute much to change detection may affect the final detection accuracy, and the amount of data is too large, which will cause the pressure of computer calculations to increase significantly. However, the band selection method proposed for change detection is still scarce.
2) *Difficulty in Extracting Change Information:* Dual-temporal images are images collected in the same scene at different times. The image radiation in different periods is different, and it is difficult to fully utilize the channel information of hyperspectral images, which increases the difficulty of detecting changed and unchanged pixels. How to make good use of the channel advantages of hyperspectral remote-sensing images is one of the challenges of change detection.
3) *Improve the Directionality of Changing Information Features:* On the one hand, the use of neural network methods for change detection mostly does not consider the features of multiple levels. On the other hand, the features extracted from the convolutional network are usually imported into the fully connected layer. The current layer of the fully connected layer will connect all the nodes of the previous layer, and weighted and accumulated accordingly. However, there is still no certainty that the feature that the network naturally weights all nodes at the previous level is most conducive to change detection, so how to improve the directionality of the change information feature is a difficult problem to be solved.

With respect to the problem of hyperspectral band redundancy, the difficulty of extracting change information from the dual-temporal image, and the problem of how to enhance the directionality of change information features, the main contributions of this article are as follows.

1) Based on slow feature analysis (SFA) theory [19], a slow-fast band selection (SFBS) method was developed. To solve the problem of band redundancy in hyperspectral images, the proposed SFBS makes it possible to select the bands in the change detection technology, while also effectively improving the accuracy of change detection.
2) This article describes a method for constructing the difference matrix. Using the channel advantages of hyperspectral images, a set of pixel pairs are expanded into a difference image, enriching the level of change information, providing more change features for change detection, and facilitating the extraction of change information features.
3) A feature fusion grouping (FFG) method is proposed. The extracted change information features are fused and then grouped, and the loss function of the network is designed to maximize the intragroup correlation and minimize the intergroup correlation, to generate a more distinguishable feature group. FFG improves the directionality of fusion features, making it more conducive to the classification of change detection.

The rest of this article is organized as follows. Section II summarizes related work in the field of change detection. Section III introduces the details of the SFBS-FFGNET. Section IV designs experiments to verify the effectiveness of the proposed method. Section V draws the conclusion of this article.

II. Related Work

There are many existing remote-sensing image change detection methods. In this article, we divide the most common change detection methods into two categories: traditional change detection methods and deep learning change detection methods. The traditional classical change detection methods are divided into two types according to whether they need to be classified as standards. One type is direct comparison methods, and the other is postclassification comparison methods.

A. Traditional Change Detection

Commonly used direct comparison methods in change detection include image difference method [20], image ratio method, vegetation index method [21], image regression

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method, mean or image logarithmic ratio method [22], change vector analysis (CVA) method [23], orthogonal transformation method [24], canonical correlation analysis (CCA) [25], etc. These methods do not have complex operations, but have high requirements for image preprocessing, and a highly reliable preprocessing link is necessary to ensure the final change detection accuracy. The accuracy of change detection based on the postclassification comparison method is related to the classification algorithm. Gong *et al.* [26] proposed a change detection method based on an image fusion strategy and a new fuzzy algorithm, which generates a difference image using complementary information from the average ratio image and the log ratio image. Celik [27] proposed in 2009 to use the principal component analysis (PCA) algorithm to generate feature vector space, and use Euclidean distance to perform K-Means clustering method for change detection. Nielsen [28] proposed an iteratively reweighted multivariate alteration detection (IR-MAD) algorithm based on MAD [29], which extends MAD by iterating the weights of different observations. Rawashdeh [30] used the method of pixel-by-pixel differential change detection to identify and evaluate newly built irrigation areas. Zhang *et al.* [31] proposed a hyperspectral collaborative sparse and nonlocal low-rank tensor change detection method, which determines whether the tensor block has changed according to the residuals of the tensor in different categories based on the representation coefficients. Tang *et al.* [32] used the pixel ratio method to determine the range of change, and determined the type of change in the classification map by comparing the classification based on the object.

B. Deep Learning Change Detection

The method based on deep learning is currently the most commonly used change detection method. Gu *et al.* [33] combined FlowNet and U-Net to propose a FlowS-Unet network model, which integrates the results of the current layer into the next layer to achieve change detection. Li *et al.* [34] obtained the weighted difference image by designing a new weighting function, which combined with the dual-temporal original image to form a three-channel color difference image and used it for the U-Net network for detection, successfully improving the detection accuracy. Huang *et al.* [35] proposed a multispectral remote-sensing image change detection method based on tensor and deep learning, which has good change detection accuracy and automation level. Li *et al.* [36] proposed a nonsmooth nonnegative matrix factorization (nsNMF) network based on synthetic aperture radar (SAR) image change detection. Wang *et al.* [37] proposed a general end-to-end two-dimensional CNN (GETNET) framework method, which constructs an affinity matrix by unmixing linear and nonlinear spectra, fusing richer features, then combining with CNNs, and finally outputting the final change detection binary map through a classifier. Du *et al.* [38] proposed a deep SFA (DSFA) method, based on the principle of SFA to suppress invariant components, highlighting the changed pixels in the image pair to achieve the effect of change detection.

Each node of a layer of a fully connected network will be connected to all of the nodes of the previous layer, and use the corresponding weights to merge the features of the previous layer. However, it is still not certain that the natural fusion by the network of the features obtained by connecting all the nodes of the previous level is most conducive to transformation detection. Therefore, it is necessary to improve the directionality of the change information feature to make it easier to distinguish, and more conducive to change detection.

III. METHODOLOGY

To address the problems and challenges mentioned above, and improve the accuracy of change detection and the robustness of dual-temporal hyperspectral images, we propose SFBS-FFGNET. The overall design scheme is shown in Fig. 1.

In the change detection method, T_1 and T_2 represent dual-temporal hyperspectral remote-sensing images. The SFBS method is used to extract the band sets that contribute most to change detection in T_1 and T_2. A method of constructing a difference matrix is then adopted, to make full use of the channel advantage of hyperspectral images. A custom convolutional network layer is then used to extract and cascade the change information feature map of the difference matrix. The output of the feature cascade is connected to the fully connected layer, and all nodes of the fully connected layer are divided into k groups, each of which contains m nodes, so that a feature group that is more directional and more conducive to change detection is generated. After generating the grouping, a softmax classifier is connected, and finally a change detection binary image is obtained.

A. Slow-Fast Band Selection

There are a large number of hyperspectral remote-sensing image bands, which provide large amounts of information for change detection, but which inevitably increase the difficulty of data processing and the computational load. There are different types of band selection methods used in other applications of hyperspectral remote-sensing images, but in the field of change detection of dual-temporal hyperspectral images, the optimal method for change detection of band selection is still unclear. Based on SFA theory and the analysis of the characteristics of the hyperspectral band, we developed an SFBS method to find the slow and fast characteristics of the hyperspectral band. The SFBS method extracts slow and fast changing bands from dual-temporal hyperspectral remote-sensing images, to form a new band set. The slow feature band can more effectively extract the features of unchanged pixels, and the fast feature band can more effectively extract the features of changed pixels. As shown in Fig. 2, in the River dataset (see Section IV-B), we have selected some slow/fast feature bands as examples. It can be seen from Fig. 2 that the slow feature band fluctuates slowly, and the band difference image obtained fluctuates slowly, clearly highlighting the unchanged area in the image. The fast feature band fluctuates faster, and the band difference image obtained fluctuates faster, clearly highlighting the changed area in the image.

Given a temporal signal $x(t) = [x_1(t), x_2(t), \ldots, x_m(t)]$ with a dimension of m, SFA aims to find a set of transformation functions $[p_1(x), p_2(x), \ldots, p_m(x)]$ and generate

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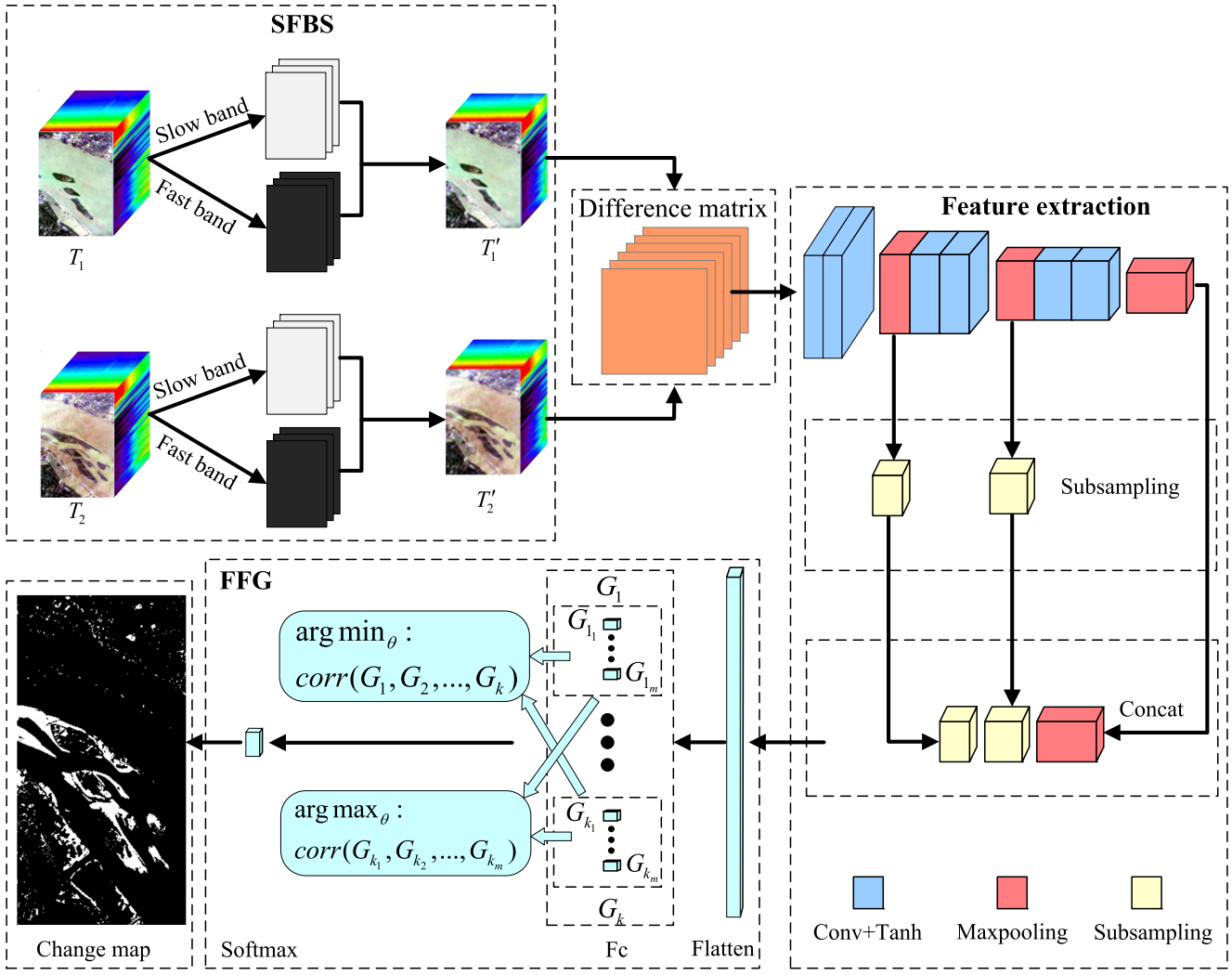


Fig. 1. Schematic of SFBS-FFGNET overall design.

a series of slow features $f(t) = [f_1(t), f_2(t), \ldots, f_m(t)] = [p_1(x), p_2(x), \ldots, p_m(x)]$. Generally speaking, we believe that some of the slowest changes among the extracted features can best represent the unchanging characteristics. The objective function of SFA can be defined as

$$\min(f_j): \langle \dot{f}_j^2 \rangle_t, \quad j \in [1, 2, \ldots, m]. \tag{1}$$

In (1), $\langle \cdot \rangle_t$ represents the expectation of the sequence, $\dot{f}$ represents the first-order timing difference $\dot{f}(t) = f(t) - f(t-1)$, and ":" indicates that $\min(f_j)$ is the expectation of minimizing the first-order timing difference of the transformed signal. The objective function should meet the following constraints:

$$\langle f_j \rangle_t = 0 \tag{2}$$

$$\langle f_j^2 \rangle_t = 1 \tag{3}$$

$$\forall i < j, \quad \langle f_i f_j \rangle_t = 0. \tag{4}$$

Equation (2) is used to standardize the data, (3) calculates the unit variance, and (4) is used to remove the correlations between the output signals. Under linear conditions, the transforming equation could be expressed as a transforming matrix, and then the original data $x(t)$ can be transformed into

$$f(t) = w^T x(t). \tag{5}$$

w in (5) represents the mapping matrix. The objective function and constraint conditions of SFA can be rewritten as follows:

$$\langle \dot{f}_j^2 \rangle_t = \langle (w_j^T \dot{x})(w_j^T \dot{x}) \rangle_t = w_j^T \langle \dot{x}\dot{x}^T \rangle_t w_j \tag{6}$$

$$\langle f_j \rangle_t = \langle w_j^T x \rangle_t = 0 \tag{7}$$

$$\langle f_j^2 \rangle_t = \langle (w_j^T x)(w_j^T x) \rangle_t = w_j^T \langle x x^T \rangle_t w_j = 1 \tag{8}$$

$$\langle f_i f_j \rangle_t = \langle (w_i^T x)(w_j^T x) \rangle_t = w_i^T \langle x x^T \rangle_t w_j = 0. \tag{9}$$

In (6), let $\langle \dot{x}\dot{x}^T \rangle_t = P$, and P represents the expectation of the first-order forward difference covariance matrix of the original data. In (8) and (9), let $\langle x x^T \rangle_t = Q$, and Q represents the expectation of the original data covariance matrix. In the dual-temporal hyperspectral image, assuming that the image data of the two temporals are T_1 and T_2, respectively, the corresponding P and Q can be expressed as

$$P = \frac{1}{n}(T_1 - T_2)(T_1 - T_2)^T \tag{10}$$

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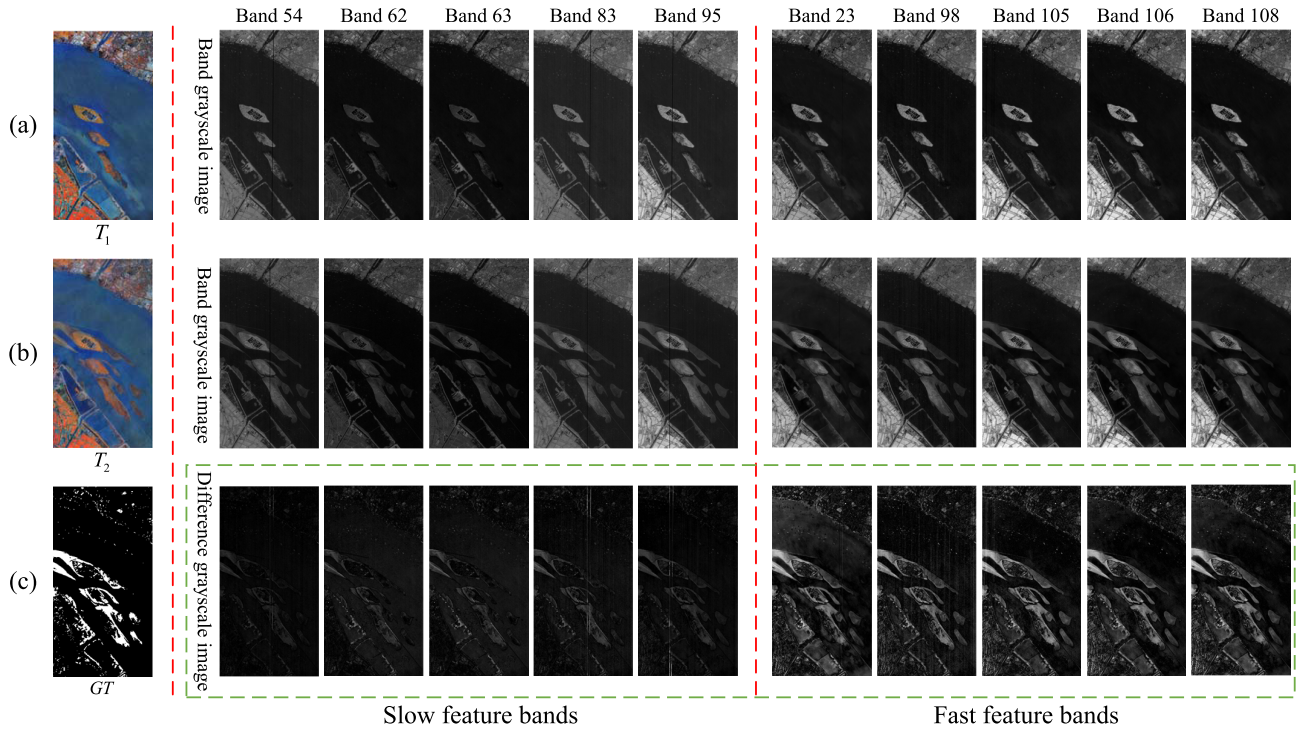


Fig. 2. Some examples of slow/fast feature bands in the River dataset. (a) Pseudo-color image of T_1 and the grayscale image of the example bands. (b) Pseudo-color image of T_2 and the grayscale image of the example bands. (c) Ground-truth map and the grayscale image of difference image between T_1 and T_2.

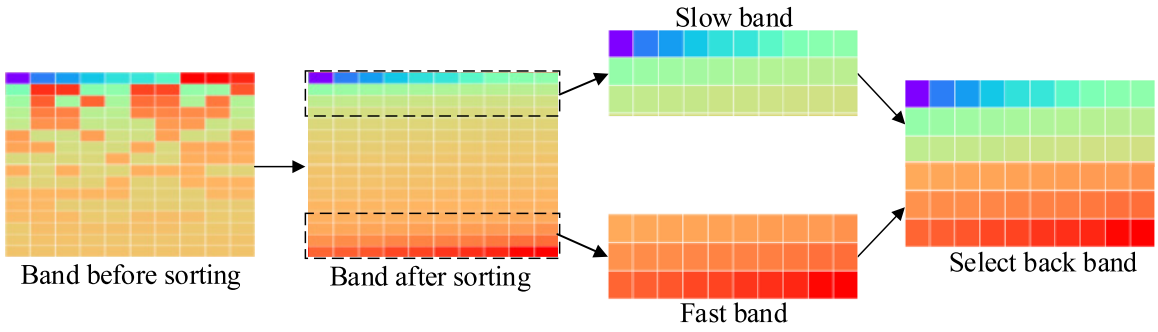


Fig. 3. Schematic of SFBS principle.

$$Q = \frac{1}{2n}\left[T_1 T_1{}^T + T_2 T_2{}^T\right]. \tag{11}$$

From (6) and (8), the objective function can be rewritten as

$$\left\langle \dot{f}_j^2 \right\rangle_t = w_j^T \left\langle \dot{x}\dot{x}^T \right\rangle_t w_j = \frac{w_j^T \left\langle \dot{x}\dot{x}^T \right\rangle_t w_j}{w_j^T \left\langle x x^T \right\rangle_t w_j} = \frac{w_j^T P w_j}{w_j^T Q w_j}. \tag{12}$$

Equation (12) can be transformed into a generalized eigenvalue solution, and the specific expression is as follows:

$$PW = QW\Lambda. \tag{13}$$

In (13), W obtained by solving is the eigenvector matrix and Λ is the eigenvalue matrix. As shown in Fig. 3, in the hyperspectral image with the number of bands n, the number of elements of the diagonal matrix Λ is also n, and each eigenvalue corresponds to the band one by one. The set of all bands of the hyperspectral image is defined as $\Omega = \{b_i\}_{i=1}^n$. Assuming that the slow characteristic band selected by SFBS is set B_s, and the selected fast characteristic band is set B_f, where B_s contains k bands and B_f contains l bands, then the constraint relationship of B_s, B_f, k, l is

$$k + l < n \tag{14}$$

$$B_s \cap B_f = \emptyset. \tag{15}$$

According to the SFA theory, the eigenvalue corresponding to the slower-changing band is smaller. An index i is found by

$$i = \arg \min_{1 \le i \le n} \{\Lambda_i\}. \tag{16}$$

The selected i is the band index corresponding to the smallest eigenvalue, the slowest characteristic band is taken

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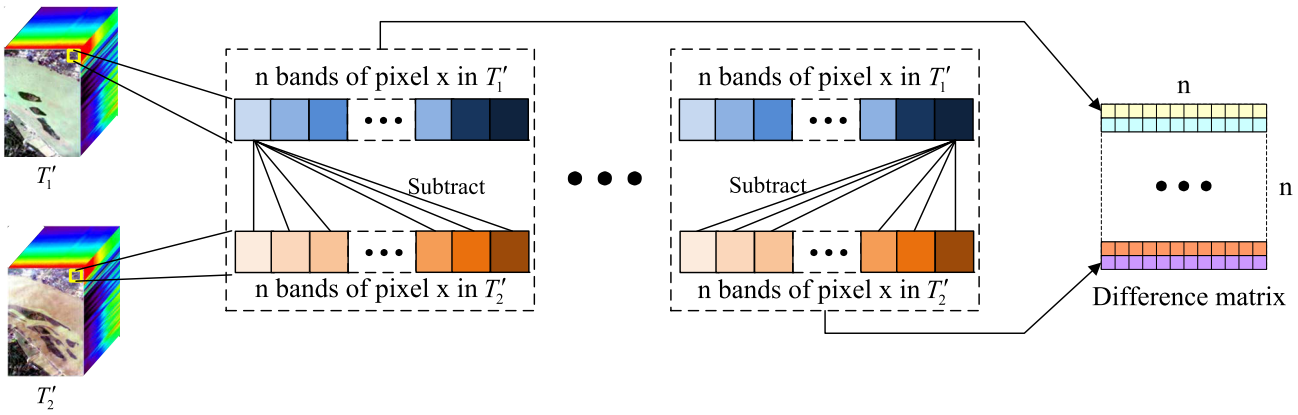


Fig. 4. Schematic of difference matrix.

out through this index, and this band is taken as the first band of the slow characteristic band set; that is, $a_1 = b_i$. In the same way, the smallest k bands are selected as the slow characteristic band set $B_s = \{a_1, a_2, \ldots, a_{k-1}, a_k\}$.

Conversely, the higher the characteristic values correspond to the fast-changing band. An index j is found by

$$j = \arg \max_{1 \le j \le n} \{\Lambda_j\}. \tag{17}$$

The selected j is the band index corresponding to the largest eigenvalue, the fastest characteristic band is taken out through this index, and this band is taken as the first band of the fast characteristic band set; that is, $c_1 = b_j$. In the same way, the largest l bands are selected as the fast characteristic band set $B_f = \{c_1, c_2, \ldots, c_{l-1}, c_l\}$.

B_s and B_f are combined as the final band set selected by SFBS

$$B = \{B_s, B_f\} = \{a_1, a_2, \ldots, a_{k-1}, a_k, c_1, c_2, \ldots, c_{l-1}, c_l\}. \tag{18}$$

B is a new set composed of two band sets selected by SFBS, and the total number of bands is $k+l$. The selection of k and l in specific experiments can be adjusted as appropriate for different datasets.

B. Design of the Difference Matrix

The core of remote-sensing image change detection is the obtaining of fused differential images. Although many methods use deep networks, it is still unavoidable to make a difference to the dual-temporal hyperspectral image at a certain step. To take advantage of the number of channels in dual-temporal hyperspectral images, we developed a method for constructing a difference matrix. Channel polling between pixel pairs is used for subtraction to mine the cross-channel gradient information between the hyperspectral image bands, and a group of pixel pairs is expanded into a difference image. This method can increase the amount of change information and provide more change characteristics for change detection. The equation for the construction of the difference matrix is as follows:

$$D_{ij} = X_{1_i} - X_{2_j}, \quad i, j \in [1, 2, 3, \ldots, n]. \tag{19}$$

Suppose the dual-temporal hyperspectral images processed by SFBS are, respectively, T_1' and T_2'. In (19), X represents a pixel arbitrarily selected in the dual-temporal hyperspectral image, X_1 represents the X pixel in T_1', i represents the ith band, X_2 represents the X pixel in T_2', j is the jth band, and n is the dimension of the data. Both i and j take arbitrary values from 1 to n; that is, the pixel pair X can construct a D matrix with a size of $n \times n$. For a hyperspectral image with a total number of pixels of M, a difference matrix of size $M \times n \times n$ can be constructed.

The design block diagram of the difference matrix is shown in Fig. 4. Taking any set of pixel pairs x from T_1' and T_2', the value vectors of the n bands of the corresponding pixels can be extracted. Let the value of the first band of the pixel in T_1' subtract the values of the n bands in T_2' in order, and the n values obtained generate the first row of the difference matrix. In the same way, the value of the nth band of the pixel in T_1' subtract the values of the n bands in T_2' in order, and the n values obtained are used as the nth row of the difference matrix, and the resulting matrix is the designed difference matrix.

C. Change Information Feature Extraction and Fusion Grouping

The difference matrix contains a wealth of channel gradient and change information, and it is very important to adopt an effective feature extraction scheme. In deep learning, the classic feature extraction network architecture used is the CNN. To better integrate the changing information features of different depths, we used a cascade of low, medium, and high features to obtain more comprehensive features. The features extracted by the convolutional network are usually imported into the fully connected layer. Each layer of a fully connected network connects all of the nodes of the previous layer, which are weighted and accumulated accordingly, but the result of the natural weighting calculation of the network is not necessarily the most conducive to transformation detection [39]. In this work, an FFG method was designed to fuse features in a direction that is beneficial to change detection, and a custom loss function was designed.

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TABLE I
ARCHITECTURE DETAILS FOR SFBS-FFGNET

Layers	Type	Ch	Kernel
LSConv1	LSConvolution+BN	32	3×3
	Activation(tanh)		
LSConv2	LSConvolution+BN	32	3×3
	Activation(tanh)		
MaxPool1	MaxPooling	-	2×2
LSConv3	LSConvolution+BN	64	3×3
	Activation(tanh)		
LSConv4	LSConvolution+BN	64	3×3
	Activation(tanh)		
MaxPool2	MaxPooling	-	2×2
LSConv5	LSConvolution+BN	128	3×3
	Activation(tanh)		
LSConv6	LSConvolution+BN	128	3×3
	Activation(tanh)		
MaxPool3	MaxPooling	-	2×2
LSConcat	MaxPool1+SubSamp1	-	4×4
	MaxPool2+SubSamp2	-	2×2
	MaxPool3	-	2×2
FC1	Fully Connected+BN	224	-
	Activation(tanh)		
FC2	Fully Connected	2	-

1) Feature Extraction: The self-defined CNN framework is shown in the feature extraction module in Fig. 1. First, the difference matrix is imported into the convolutional layer with 32 channels, and then the features here are taken out as low-level features through a maximum pooling layer. Then, through a convolutional layer with 64 channels, the pooled features are extracted as intermediate features. The third level is connected to a convolutional layer with 128 channels, and the pooled features are extracted as high-level features. The convolutional layers in the network are all 3×3 convolution kernels, and the kernels size of the pooling layer is 2×2. Because the size of the feature map extracted by the first two levels does not match that of the last level, in the case of the least loss of feature information, the features extracted by the first two levels are downsampled to make the size of the three levels' feature map consistent. The specific network details are shown in Table I. The low, medium, and high features are cascaded together as the extracted change information feature.

2) Feature Fusion Grouping: For the features extracted by the convolutional network, to generate features that are more conducive to change detection, we developed an FFG method.

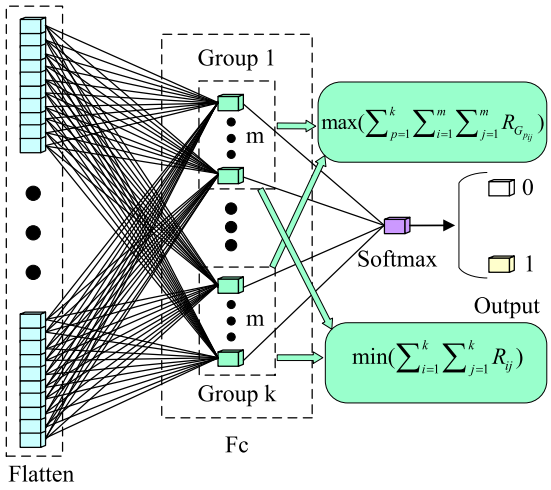


Fig. 5. Schematic of FFG.

Fusion here means that a node merges the information from all the upper nodes. Because the naturally fused information is not necessarily the most conducive to change detection, the feature groups of multiple feature sets help to enhance the separability of the fused features, so an FFG method is used. As shown in Fig. 5, all nodes in the current layer of the fully connected layer network are divided into k groups, each containing m nodes. By designing a grouping loss function to maximize intragroup correlation and minimize intergroup correlation [40], a feature group with more discriminative features can be generated, so that the softmax classifier can perform better, which is the core of the whole method.

Assuming that the output layer is the nth layer of the network, in the design of this network, the output can be expressed as

$$\begin{aligned} f_n(x) &= f(\theta, x) = s_2(w_n f_{n-1}(x) + b_n) \\ &= s_2(w_n s_1(w_{n-1} f_{n-2}(x) + b_{n-1}) + b_n). \end{aligned} \tag{20}$$

In (20), x is the network input, $f_n(x)$ represents the output of the nth layer, θ is the global network parameter, and w_n and b_n represent the weight matrix and deviation vector of the nth layer, respectively. In the method, in this article, $s_1(\cdot)$ represents the activation function tanh, and $s_2(\cdot)$ represents the activation function softmax.

Grouping: In FFG, the set of all nodes is defined as G, and is divided into k groups, each of which contains m nodes, so $G = [G_1, G_2, \ldots, G_k]$. The expression of each group is as follows:

$$G_p = \left[x_{p_1}, x_{p_2}, \ldots, x_{p_m}\right], \quad p \in [1, 2, \ldots, k]. \tag{21}$$

In (21), x_{p_m} represents the mth node of the pth group. Grouping needs to meet the following constraints:

$$\begin{cases} \bigcup_{i=1}^{k} G_i = G \\ \bigcap_{i=1}^{k} G_i = \emptyset. \end{cases} \tag{22}$$

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Equation (22) indicates that the union of all groups must be equal to the entire set, and there is no intersection on nodes between groups, to avoid the same node from appearing in different groups.

Loss Function: For any group G_p, $p \in [1, 2, \ldots, k]$, to maximize the intragroup correlation of the group, we first calculate the covariance matrix of m nodes in the group. The specific equation is

$$C_p = \frac{1}{m}\sum_{i=1}^{m}\left(x_{p_i} - \bar{x}_p\right)\left(x_{p_i} - \bar{x}_p\right)^T, \quad p \in [1, 2, \ldots, k] \tag{23}$$

where C_p is the covariance matrix in the pth grouping group, and $\bar{x}_p = (1/m)\sum_{i=1}^{m} x_{p_i}$ is the mean value of all elements in the pth grouping. Then we calculate the corresponding correlation coefficient matrix as

$$R_{p_{ij}} = \frac{\left|C_{p_{ij}}\right|}{\sqrt{C_{p_{ii}} C_{p_{jj}}}}, \quad p \in [1, 2, \ldots, k]. \tag{24}$$

As shown in Fig. 5, the intragroup correlation depends on the sum of k intragroup correlations, and the objective function for maximizing intragroup correlation can be defined as

$$A = \max\left(\sum_{p=1}^{k}\sum_{i=1}^{m}\sum_{j=1}^{m} R_{p_{ij}}\right). \tag{25}$$

Considering that the loss function in the network can only be trained toward a smaller loss value, the loss function that maximizes the correlation within the group is designed as

$$L_{11} = 1 - \frac{A}{km^2}. \tag{26}$$

In (26), L_{11} is the first part of the loss function of the entire fusion grouping method. We then normalize A and subtract it from 1.

To minimize the correlation between groups, let $y_p = (1/m)\sum_{i=1}^{m} x_{p_i}$ represent the pth group, where $p \in [1, 2, \ldots, k]$. Similarly, the equation for calculating the covariance matrix of k groups is

$$C = \frac{1}{k}\sum_{p=1}^{k}\left(y_p - \bar{y}\right)\left(y_p - \bar{y}\right)^T. \tag{27}$$

In (27), C is the coordinated difference matrix between groups, and $\bar{y} = (1/k)\sum_{p=1}^{k} y_p$ is the mean value of all elements in the pth groups. The corresponding correlation coefficient matrix between groups can be calculated as

$$R_{ij} = \frac{\left|C_{ij}\right|}{\sqrt{C_{ii} C_{jj}}} \tag{28}$$

where R represents the correlation coefficient matrix between groups and R_{ij} represents the value of the corresponding position of the correlation coefficient matrix. As shown in Fig. 5, the correlation between groups depends on the correlation between k groups, so the objective function to minimize the correlation between groups can be defined as

$$B = \min\left(\sum_{i=1}^{k}\sum_{j=1}^{k} R_{ij}\right). \tag{29}$$

Similar to (26), the loss function of the second part of FFG is designed as follows:

$$L_{12} = \frac{B}{k^2}. \tag{30}$$

L_{12} is obtained by normalizing B. Considering that the idea of the whole method is to maximize the correlation within the group and minimize the correlation between the groups, to generate a more discriminative fusion feature group, we enhance the directionality of the group feature, and facilitate the judgment of the softmax classifier. So the total loss function of this part is defined as

$$L_1 = \lambda_1 * L_{11} + \lambda_2 * L_{12} \tag{31}$$

where L_1 is the loss function of FFG. The coefficients of the two parts of the loss function are λ_1 and λ_2, respectively, and the values of the two parameters can be adjusted as needed.

Categorical cross-entropy is usually used as a classification loss function. It can be used to evaluate the difference between the current training probability distribution and the true distribution. The smaller the distance between the actual output and the expected output, the smaller the corresponding cross-entropy value, and the closer the probability distribution. Categorical cross-entropy is usually used in combination with softmax to achieve the effect of two classifications. The categorical cross-entropy equation is as follows:

$$L_{\text{cct}} = -[y \ln x + (1 - y)\ln(1 - x)]. \tag{32}$$

In (32), y represents the true label of the pixel, and x represents the actual output of the current neural network. Categorical cross-entropy needs to be used in combination with softmax to achieve two classifications. The definition expression of softmax is

$$\sigma(z)_j = \frac{e^{z_j}}{\sum_{k=1}^{K} e^{z_k}}. \tag{33}$$

In the following, $\sigma(z)_j$ is uniformly expressed as σ. After combining categorical cross-entropy and softmax, the loss function expression of this part is obtained

$$L_2 = -[y \ln(\sigma(x)) + (1 - y)\ln(\sigma(1 - x))]. \tag{34}$$

According to (31) and (34), the loss function of the entire network is

$$L = \lambda_3 * L_1 + \lambda_4 * L_2. \tag{35}$$

In (35), λ_3, and λ_4 represent the weight coefficients of the two parts of L_1 and L_2, and these two coefficients should be determined according to the specific situation.

IV. Experiments

In this section, we conduct sufficient experiments on three real hyperspectral datasets to verify the effectiveness of SFBS-FFGNET. The first is the River dataset collected from a river in Jiangsu Province, China, the second is the Farmland dataset collected in Yancheng, Jiangsu Province, China, and the third is the Hermiston dataset collected in the urban scene of Hermiston. This experiment selected three classic change

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detection methods and two newer methods as comparison methods. This article also designed two ablation experiments for SFBS-FFGNET to verify the effectiveness of SFBS and FFG. In the experiment, we established a comprehensive index system to evaluate the results of change detection, and conducted an objective analysis of the effect of the experiment.

A. Introduction to Experimental Methods

In SFBS-FFGNET, the specific network details are shown in Table I. In the experiment, two types of methods are compared with SFBS-FFGNET. The traditional methods include CVA [41], PCA-CVA [42], patch-based CNN [12] method. Deep learning methods include DSFA proposed by Du *et al.* [38] and GETNET proposed by Wang *et al.* [37]. In SFBS-FFGNET, an SFBS method and an FFG method are combined. To verify the effectiveness of the two methods, we designed two ablation experiments. These experiments involved keeping the other conditions of the method unchanged, removing the SFBS module, experimenting on different datasets, and comparing the experimental results with the complete SFBS-FFGNET. Then, keeping the other conditions of the method unchanged, the FFG module and the corresponding loss function was removed, experiments were performed on different datasets, and the experimental results were compared with the complete SFBS-FFGNET.

To better evaluate the results of the experiment, this experiment used a variety of evaluation indicators, including OA, OA of changed pixels (OA_CH), OA of unchanged pixels (OA_UN) and Kappa coefficient. OA reflects the total accuracy of the experimental results, and is the most convincing and intuitive indicator. OA_CH reflects the recognition accuracy of the changed pixels, and OA_UN reflects the detection accuracy of the unchanged pixels. The Kappa coefficient is used for consistency testing. It can objectively reflect the consistency of the change detection map and the ground-truth map. To reduce the randomness of the experimental data, we experimented with each method five times and took the average and standard deviation as the final change detection data. In addition, we select training samples based on the principle that the ratio of positive samples (changed pixels) and negative samples (unchanged pixels) is 1:2 [37].

B. Experiment on River Dataset

The River dataset [37] was collected from a river in Jiangsu Province, China. The two hyperspectral images were acquired on May 3, 2013, and December 31, 2013. The spectral range of the River dataset is 0.4–2.5 μm, the spectral resolution is 10 nm, its spatial resolution is 30 m, and there are 242 spectral bands in total. The image size in the dataset is 463 $\times$ 241 pixels, and 198 bands are available after removing the noise. Fig. 6 shows dual-temporal pseudo-color map and ground-truth map of the River dataset. On the ground-truth map, the changed pixels are white blocks, and the unchanged pixels are black blocks.

The River dataset has a total of 111 583 pixels, of which a total of 9698 pixels are changed, and a total of 101 885 pixels are unchanged. Fig. 7 shows the spectra of unchanged pixels and changed pixels, where (a) represents the spectral curves of the unchanged pixels at different times, and (b) represents the spectral curves of the changed pixels at different times. In the experiment, 1250 high-confidence changed pixels and 2500 unchanged pixels were selected as training samples in the CVA predetection results, accounting for 3.37% of the total pixels.

Table II shows the result of testing all methods on the River dataset. SFBS-FFGNET achieved the best performance on OA and Kappa. Among the other methods, PCA-CVA had the highest detection accuracy on OA_CH, but its unchanged pixel

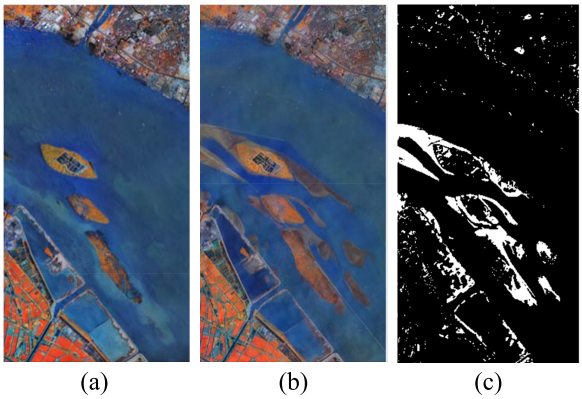


Fig. 6. Dual-temporal imagery of the River dataset obtained in (a) May 2013, (b) December 2013, and (c) ground-truth map.

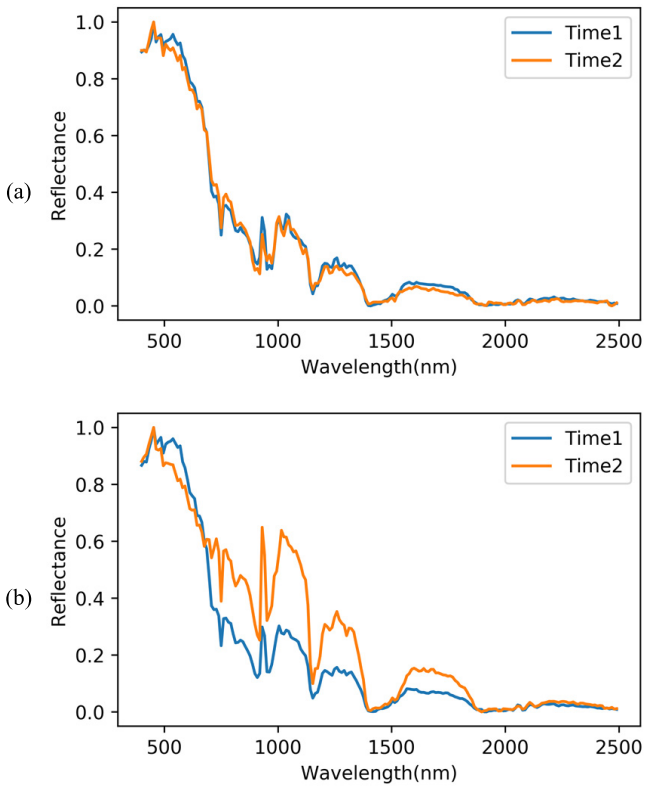


Fig. 7. Example spectra of unchanged pixels and changed pixels in the River dataset. (a) Unchanged pixel spectrum curve. (b) Changed pixel spectrum curve.

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TABLE II
CHANGE DETECTION RESULTS OF THE RIVER DATASET (TRAINING SAMPLES ACCOUNTED FOR 3.37%, REPEATED RUN FIVE TIMES)

Different methods	Evaluation index			
	OA_CH	OA_UN	OA	Kappa
CVA	0.861	0.892	0.889	0.520
PCA-CVA	**0.950**	0.952	0.952	0.748
CNN	0.804±0.027	0.945±0.003	0.933±0.002	0.640±0.010
DSFA	0.688±0.013	0.981±0.007	0.948±0.005	0.721±0.009
GETNET	0.433±0.018	**0.998±0.001**	0.949±0.001	0.574±0.015
SFBS-FFGNET(without FSBS)	0.535±0.048	0.996±0.001	0.956±0.003	0.655±0.037
SFBS-FFGNET(without FFG)	0.712±0.082	0.982±0.006	0.959±0.002	0.731±0.032
SFBS-FFGNET(with FSBS, with FFG)	0.713±0.013	0.991±0.002	**0.967±0.001**	**0.771±0.004**

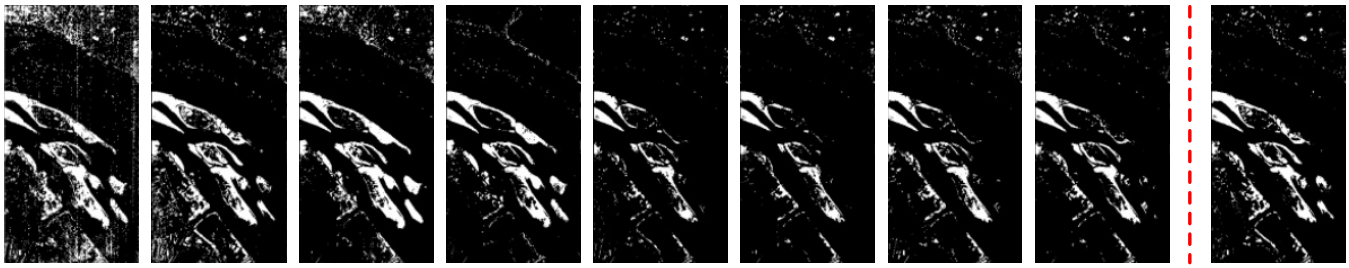


Fig. 8. Change detection maps of different methods on River dataset. (a) CVA. (b) PCA-CVA. (c) CNN. (d) DSFA. (e) GETNET. (f) SFBS-FFGNET (without SFBS). (g) SFBS-FFGNET (without FFG). (h) SFBS-FFGNET. (i) Ground-truth map.

detection accuracy OA_UN was more general, so the OA was lower than that of SFBS-FFGNET. GETNET had the highest OA_UN, but its OA_CH was very low, so its OA and Kappa were not as good as those of SFBS-FFGNET. SFBS-FFGNET was far ahead of the comparative DSFA method in all aspects.

SFBS-FFGNET had a higher OA and Kappa than SFBS-FFGNET without SFBS, and OA_UN was slightly lower than SFBS-FFGNET without SFBS, indicating that SFBS can improve the robustness of the method and help improve the OA. SFBS-FFGNET's OA_CH was slightly lower than that of SFBS-FFGNET without FFG, and OA_UN was higher than that of SFBS-FFGNET without FFG, and was also leading in OA. Overall, SFBS-FFGNET, SFBS-FFGNET without FFG, and SFBS-FFGNET without SFBS, ranked as the top three in overall performance of all methods.

Fig. 8 presents the change detection maps obtained by the different methods. It can be seen from Fig. 8 that the overall performance of our method was the closest to the ground-truth map, and the obvious change pixels in the bottom-left corner area were not detected, which is the main reason for the low OA_CH. Compared with SFBS-FFGNET, DSFA identified more unchanged pixels in the middle area of the image as changed pixels, and most of the changed pixels in the top-right corner were detected as unchanged pixels. Comparing our method with GETNET, we can clearly see that the middle area SFBS-FFGNET method OA_CH was higher. In general, SFBS-FFGNET had the best detection performance.

As shown in Fig. 8, the detection of unchanged pixels without SFBS was the best of the three methods, but the detection effect of changed pixels was very poor, and the overall detection performance was obviously not as good as the complete SFBS-FFGNET. This observation illustrates the reliability of SFBS for change detection. SFBS-FFGNET without FFG was basically the same as SFBS-FFGNET in the detection of changed pixels, but was inferior to SFBS-FFGNET in the detection of unchanged pixels, so the overall effect was slightly worse than SFBS-FFGNET. In general, the overall effects of Fig. 8(f)–(h) were stronger than those of the other methods used for comparison.

C. Experiment on Farmland Dataset

The Farmland dataset [43] is taken from farmland near Yancheng, Jiangsu Province, China. The first one was taken on May 3, 2006, and the latter was taken on April 23, 2007. The image size in the dataset is 450 × 140 pixels, and 155 bands are available after removing the noise. Fig. 9 contains dual-temporal pseudo-color map and ground-truth

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TABLE III
CHANGE DETECTION RESULTS OF THE FARMLAND DATASET (TRAINING SAMPLES ACCOUNTED FOR 20.95%, REPEATED RUN FIVE TIMES)

Different methods	Evaluation index			
	OA_CH	OA_UN	OA	Kappa
CVA	**0.981**	0.947	0.957	0.898
PCA-CVA	0.959	0.970	0.967	0.920
CNN	0.960±0.004	0.948±0.007	0.952±0.006	0.885±0.013
DSFA	0.060±0.005	0.840±0.013	0.614±0.008	-0.120±0.008
GETNET	0.960±0.013	**0.977±0.009**	0.972±0.003	0.933±0.007
SFBS-FFGNET(without FSBS)	0.980±0.002	0.968±0.003	0.971±0.001	0.932±0.002
SFBS-FFGNET(without FFG)	0.979±0.002	0.970±0.002	0.973±0.001	0.935±0.002
SFBS-FFGNET(with FSBS, with FFG)	0.978±0.001	0.975±0.001	**0.976±0.001**	**0.943±0.001**

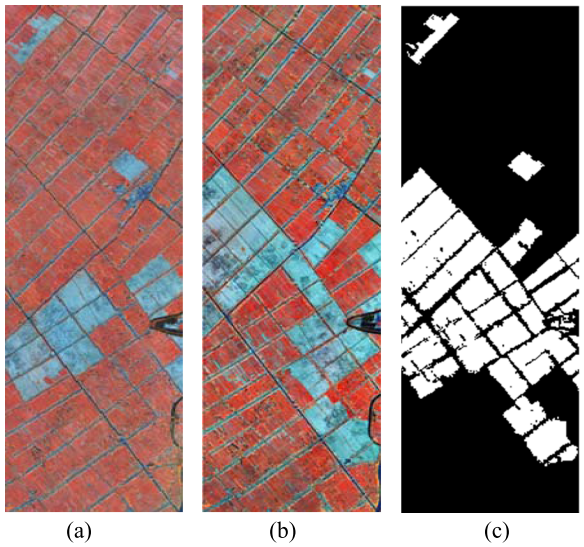


Fig. 9. Dual-temporal imagery of the Farmland dataset obtained in (a) May 3, 2006, (b) April 23, 2007, and (c) ground-truth map.

map of the Farmland dataset. On the ground-truth map, the changed pixels are marked as white, and the unchanged pixels are marked as black.

The Farmland dataset has a total of 63 000 pixels, of which a total of 18 277 pixels are changed, and a total of 44 723 pixels are unchanged. Fig. 10 shows the spectra of unchanged pixels and changed pixels, where (a) represents the spectral curves of the unchanged pixels at different times, and (b) represents the spectral curves of the changed pixels at different times. To better compare with the GETNET method, the same number of pixels as GETNET is selected as the training sample; that is, 4400 changed pixels and 8800 unchanged pixels are selected from the high-confidence pixels of the CVA predetection as the training sample. The ratio of training samples of total pixels is 20.95%.

Table III shows the results of all the experimental methods on the Farmland dataset. It can clearly be seen from Table III

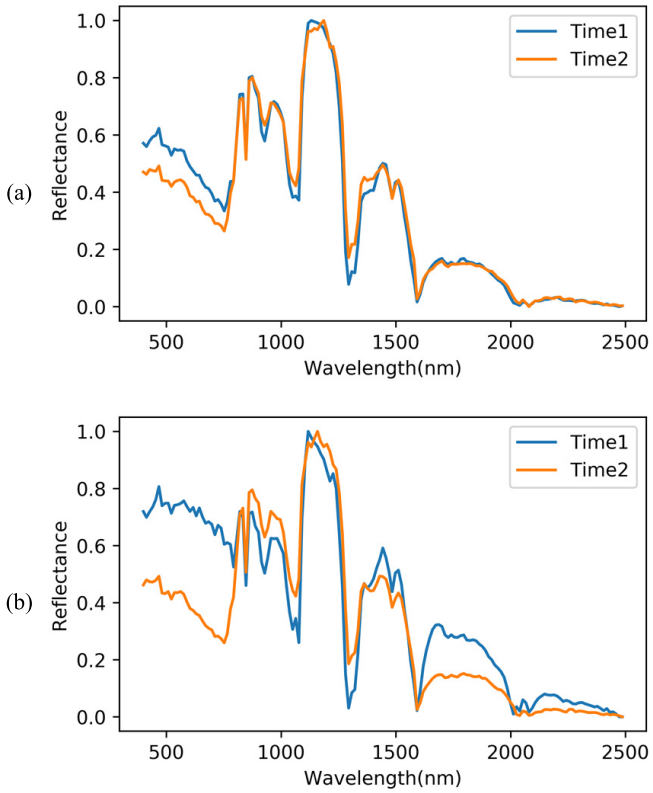


Fig. 10. Example spectra of unchanged pixels and changed pixels in the Farmland dataset. (a) Unchanged pixel spectrum curve. (b) Changed pixel spectrum curve.

that SFBS-FFGNET had the highest OA and Kappa. Although SFBS-FFGNET was not the best among the methods with respect to OA_CH and OA_UN, they were ranked third and second, and the difference in value was very small. GETNET achieved the best accuracy on OA_UN, but its OA_CH accuracy was only 96%, so the OA and Kappa were both inferior to those of SFBS-FFGNET. Although CVA's

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TABLE IV
CHANGE DETECTION RESULTS OF THE HERMISTON DATASET (TRAINING SAMPLES ACCOUNTED FOR 10%, REPEATED RUN FIVE TIMES)

Different methods	Evaluation index			
	OA_CH	OA_UN	OA	Kappa
CVA	0.532	**0.999**	0.939	0.661
PCA-CVA	**0.932**	0.986	0.979	0.906
CNN	0.760±0.014	0.975±0.020	0.947±0.019	0.760±0.070
DSFA	0.912±0.028	0.801±0.013	0.815±0.010	0.462±0.016
GETNET	0.749±0.031	0.984±0.011	0.954±0.006	0.781±0.019
SFBS-FFGNET(without FSBS)	0.893±0.004	0.995±0.003	0.982±0.003	0.916±0.013
SFBS-FFGNET(without FFG)	0.905±0.009	0.982±0.037	0.971±0.033	0.885±0.108
SFBS-FFGNET(with FSBS, with FFG)	0.915±0.004	0.995±0.001	**0.985±0.001**	**0.931±0.002**

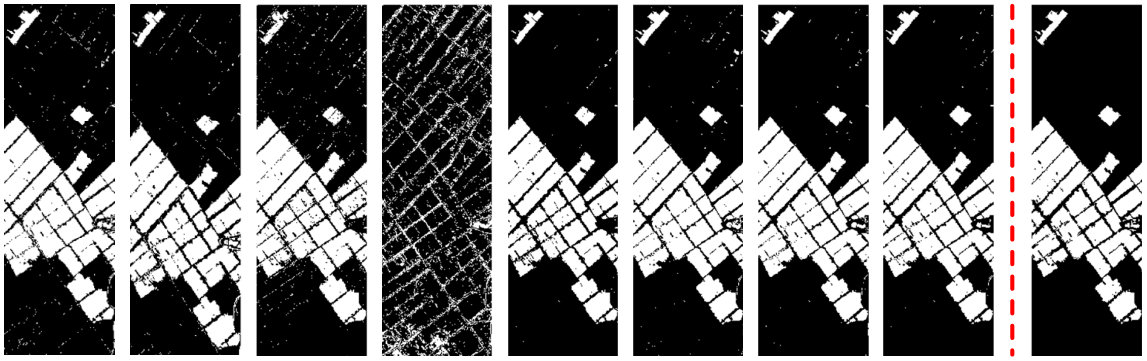


Fig. 11. Change detection maps of different methods on Farmland dataset. (a) CVA. (b) PCA-CVA. (c) CNN. (d) DSFA. (e) GETNET. (f) SFBS-FFGNET (without SFBS). (g) SFBS-FFGNET (without FFG). (h) SFBS-FFGNET. (i) Ground-truth map.

OA_CH was the highest of all methods, its OA_UN was much lower than SFBS-FFGNET, so the overall performance ranked sixth among all methods. The more complex changes in the Farmland dataset may be the reason for the poor performance of DSFA.

Analyzing the last three rows of data in Table III, SFBS-FFGNET improved OA_UN more than SFBS-FFGNET without SFBS, and SFBS-FFGNET also improved OA_UN more than SFBS-FFGNET without FFG. Due to the large proportion of constant pixels, OA was still higher than the other two methods. The above analysis and data show that both SFBS and FFG had produced performance improvements. In general, the overall effect of SFBS-FFGNET without SFBS and FFG was the same as that of GETNET, and after adding the two modules SFBS and FFG, the effect was significantly better than GETNET.

The change detection maps obtained by all the experimental methods used on the Farmland dataset is shown in Fig. 11. It can be seen from the ground-truth map that the change area of the Farmland dataset is relatively concentrated, and there are almost no scattered change pixels. From Fig. 11, the three traditional methods detected a large number of constant pixels as changing pixels, and the overall visual effect was not good. DSFA is an unsupervised deep network method, which is not necessarily suitable for datasets with higher complexity. GETNET, SFBS-FFGNET without SFBS, SFBS-FFGNET without FFG, and complete SFBS-FFGNET had similar change detection performance. The change detection map obtained by GETNET was the closest to the ground-truth map in the black area, but there was a gap in the change area on the right edge, which is why GETNET was lower on OA_CH. SFBS-FFGNET had a high degree of agreement between the black area and the white area and the truth map, and the overall performance of its change detection map was the closest to the ground-truth map.

From Fig. 11(f)–(h), SFBS-FFGNET without SFBS had more white pixel blocks in the top-right corner than the other two, while in other areas, the three algorithms were very close. On the whole, the performance of the three was very good. When our method did not include SFBS and FFG, the overall effect and accuracy was decreased to a certain extent.

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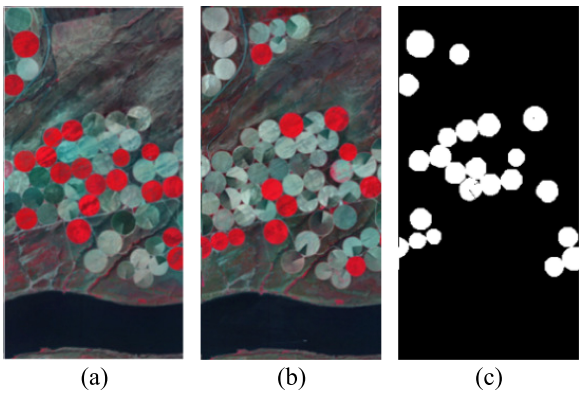


Fig. 12. Dual-temporal imagery of the Hermiston dataset obtained in (a) 2006, (b) 2007, and (c) ground-truth map.

D. Experiment on Hermiston Dataset

The Hermiston dataset was collected in the urban scene of Hermiston. The two hyperspectral images were taken in 2004 and 2007, respectively, and the sensor used in the shooting was HYPERION. The Hermiston dataset contains five major changes, its image size is 390 × 200, and 242 bands are available after removing the noise. In this article, we treat the five types of changes as changed pixels. Fig. 12 shows the pseudo-color images of the images taken in 2004 and 2007, respectively. Fig. 12(c) is the ground-truth map, in which changed pixels are marked as white, and unchanged pixels are marked as black.

The Hermiston dataset has a total of 78 000 pixels, of which a total of 9986 pixels are changed, and a total of 68 014 pixels are unchanged. Fig. 13 shows the spectra of unchanged pixels and changed pixels, where (a) represents the spectral curves of the unchanged pixels at different times, and (b) represents the spectral curves of the changed pixels at different times. Due to the variety of changes in the Hermiston dataset, 2600 high-confidence change pixels and 5200 unchanged pixels were selected as training samples in the CVA predetection results, accounting for 10% of the total pixels.

Table IV presents the detection accuracy of the different methods used in the experiment on the Hermiston dataset. PCA-CVA performed best on OA_CH, but its OA_UN was slightly lower, so OA ranked third among all methods. The OA_UN of CVA was the best among all methods, but its OA_CH was only 53.2%, which resulted in the overall detection accuracy of CVA and Kappa being very unsatisfactory. The best overall performance was that of the complete SFBS-FFGNET, with an OA of 98.5% and the highest Kappa value of 0.931. Although the OA_CH and OA_UN of SFBS-FFGNET were not the highest, OA_CH is ranked second, and OA_UN was also very close to the first.

In the last three rows of Table IV, the experimental results of SFBS-FFGNET, SFBS-FFGNET without SFBS, and SFBS-FFGNET without FFG are quite different. The OA_CH of SFBS-FFGNET without SFBS is the lowest among the three methods, but its OA_UN performance is the second best, and unchanged pixels account for a large proportion of the total pixels, so SFBS-FFGNET without SFBS has both

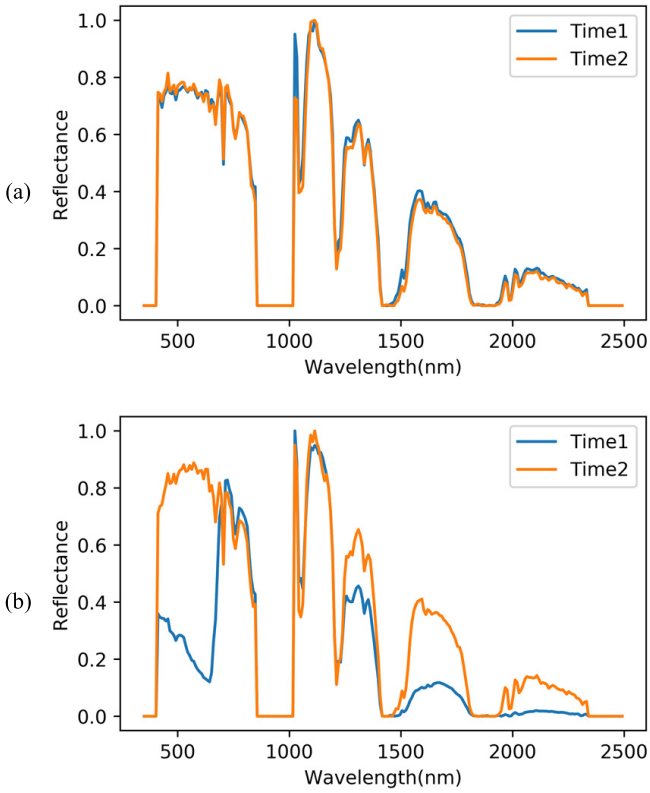


Fig. 13. Example spectra of unchanged pixels and changed pixels in the Hermiston dataset. (a) Unchanged pixel spectrum curve. (b) Changed pixel spectrum curve.

OA and Kappa performance the second best. The OA_UN of SFBS-FFGNET without FFG is the lowest among the three methods, and its OA_CH performance is the second best, but the ratio of changed pixels to the total pixels is very small, so the OA and Kappa performance of SFBS-FFGNET without FFG is the worst. SFBS-FFGNET's OA_CH is 2.2% higher than SFBS-FFGNET without SFBS, and its OA_UN is 1.3% higher than SFBS-FFGNET without FFG. In summary, SFBS-FFGNET has the best performance on both OA and Kappa, as well as the best robustness.

Fig. 14 shows a binary map of the change detection achieved by the various methods on the Hermiston dataset. In terms of overall visual effects, DSFA has the worst detection performance, and SFBS-FFGNET has the highest consistency with the ground-truth map. In the binary detection images of CVA, CNN, and GETNET, many white areas are not detected, and there are a lot of gaps in the detected white areas, which indicates that many changed pixels have been missed. In the change detection binary image of PCA-CVA, CNN, GETNET, and SFBS-FFGNET without FFG, there are many areas that should be black but are detected as white areas, which indicates that there are many false alarm pixels. From the change detection binary image of SFBS-FFGNET, it can be seen that there are still a small number of missed pixels and very few false alarm pixels. Comparing the change detection binary map of all methods with the ground-truth map, it can be clearly seen that SFBS-FFGNET is closest to the ground-truth map.

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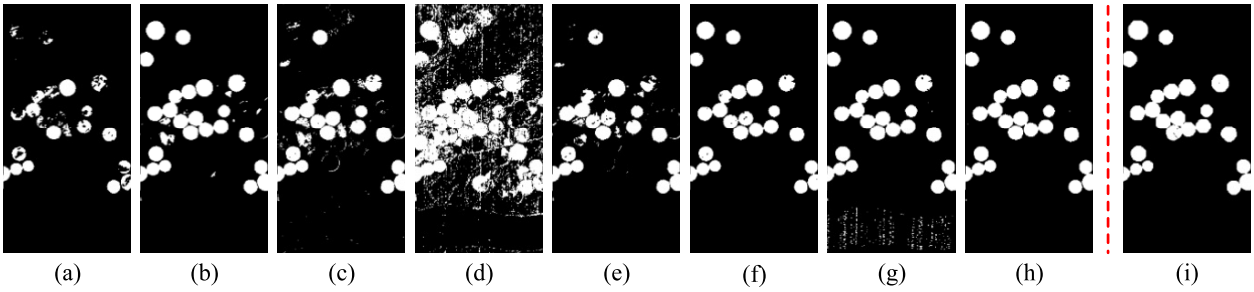


Fig. 14. Change detection maps of different methods on Hermiston dataset. (a) CVA. (b) PCA-CVA. (c) CNN. (d) DSFA. (e) GETNET. (f) SFBS-FFGNET (without SFBS). (g) SFBS-FFGNET (without FFG). (h) SFBS-FFGNET. (i) Ground-truth map.

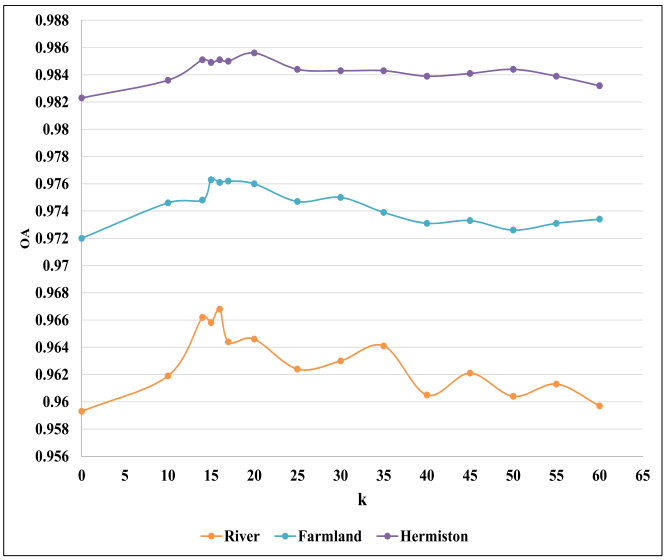


Fig. 15. Relationship between k and OA of the FFG module on the three datasets.

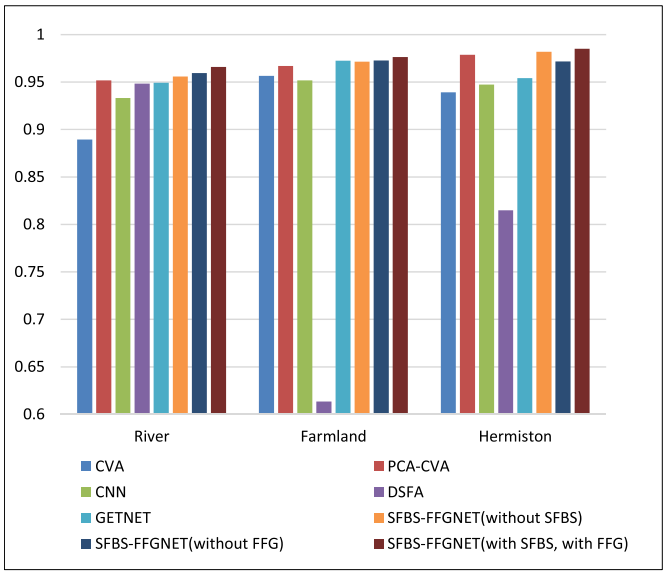


Fig. 16. OA comparisons chart of different methods on three datasets.

In Fig. 14(f)–(h), respectively, represent SFBS-FFGNET without SFBS, SFBS-FFGNET without FFG, and a complete SFBS-FFGNET change detection binary map. Compared with the complete SFBS-FFGNET, the SFBS-FFGNET without SFBS has small black patches in many white areas, which means that many changed pixels are detected as unchanged pixels. Compared with the complete SFBS-FFGNET, the change detection binary image of SFBS-FFGNET without FFG has a large number of false alarm pixels directly below. In summary, the change detection binary graph of SFBS-FFGNET is the best among the three methods.

E. Hyperparameter Analysis

In our experiment, we set the grouping number, k, of the FFG module to 16. Under the framework of this article, we conducted experiments with different grouping schemes on three datasets, and the experimental results obtained are shown in Fig. 15. In the experiment, we set the grouping interval to 5, and refined the number of groups around $k = 15$ to obtain a more precise optimal grouping scheme. It can be seen from Fig. 15 that the impact of parameter k on the accuracy of the three datasets is approximately equivalent. When the number of groups is 0 (no grouping), OA is the lowest. When the number of groups was around 15, the change detection accuracy of the three datasets was the highest, and then as the number of groups increased, the OA volatility decreased. By analyzing the optimal grouping scheme of the experiment on the three datasets, it was finally decided to set the number of groups, k, to 16. When the total number of nodes remains unchanged, the number of nodes in each group is determined by k, so no more experiments were performed on the parameter m.

F. Experiment Summary

The SFBS-FFGNET proposed in this article was used in experiments on three hyperspectral datasets: River, Farmland, and Hermiston. The OAs of the experiments were compared with those of three classic methods, CVA, PCA-CVA, and CNN; the two new methods, DSFA and GETNET; and their own two ablation methods, SFBS-FFGNET without SFBS, and SFBS-FFGNET without FFG. The comparison of the OAs of all methods on the three datasets is shown in Fig. 16. In the River dataset, SFBS-FFGNET performed best, SFBS-FFGNET (without SFBS), SFBS-FFGNET without FFG, and

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SFBS-FFGNET show an increasing trend, and the three methods were better than the other methods. In the Farmland dataset, DSFA had the lowest detection accuracy, and the best performance among traditional methods was that of PCA-CVA. The performance of GETNET was the closest to SFBS-FFGNET without SFBS, but it was still lower than that of SFBS-FFGNET without FFG and SFBS-FFGNET. The OAs of SFBS-FFGNET without SFBS, SFBS-FFGNET without FFG, and SFBS-FFGNET showed an increasing trend. In the Hermiston dataset, the best-performing traditional method was PCA-CVA. Among the methods based on deep learning, CNN and GETNET perform better, with OA both about 95%. In the ablation experiment, SFBS-FFGNET without SFBS performed better, and SFBS-FFGNET without FFG performed the worst among ablation comparison methods, but SFBS-FFGNET still performed best with OA. Regarding SFBS-FFGNET without SFBS, SFBS-FFGNET without FFG, and SFBS-FFGNET does not show an increasing trend, which is determined by the difference between the datasets. Experiments have shown that both SFBS and FFG can effectively improve the accuracy of change detection.

V. Conclusion

In this article, we proposed a change detection method named SFBS-FFGNET. In SFBS-FFGNET, based on SFA, SFBS is proposed, which selects the slow characteristic band and the fast characteristic band to extract the changed and unchanged features in the change detection, so as to better separate the changed and unchanged pixels. SFBS-FFGNET also proposed an FFG method and designed a related loss function. The main function of FFG is to generate more discriminative feature groups, which effectively improved the accuracy of change detection. Our experimental method is to select high-confidence samples from the CVA predetection results as training samples, which is a weakly supervised method. The method in this article has been experimented on three hyperspectral image datasets, and the experiment shows that the complete SFBS-FFGNET is the most stable of all methods and the best OA and Kappa. For SFBS and FFG, two ablation experiments were done in this article, and the experiments proved that the two methods have a good improvement in change detection.

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